

WAEK Mathematics 2025 Solutions

Answers and worked solutions from the Cheetah WAEK 2025 question bank.

Question 1

Find $(101_2)^2$, expressing the answer in base 2.

Answer: D

Convert 101_2 to decimal: $1 \cdot 2^2 + 0 \cdot 2 + 1 = 5$. Then $5^2 = 25$. Converting 25 to base 2 gives 11001_2 .

Question 2

If three children share ₦10.50 among themselves in the ratio 6 : 7 : 8, how much is the largest share?

Answer: B

Total parts = $6 + 7 + 8 = 21$. One part $\frac{10.50}{21}$. Largest share $\frac{8}{21} \times 10.50$.

Question 3

Express 0.000834 in standard form.

Answer: A

Move the decimal point four places to the right to get 8.34. Hence $0.000834 = 8.34 \times 10^{-4}$.

Question 4

Simplify $0.027^{-\frac{1}{3}}$.

Answer: B

$0.027 = \frac{27}{1000} = \left(\frac{3}{10}\right)^3$. Therefore $0.027^{-1/3} = \left(\frac{3}{10}\right)^{-1} = \frac{10}{3} = 3\frac{1}{3}$.

Question 5

Given that $\log_2 a = \log_8 4$, find a .

Answer: B

Let $\log_8 4 = y$. Then $8^y = 4$, so $2^{3y} = 2^2$ and $y = \frac{2}{3}$. Hence $\log_2 a = \frac{2}{3}$, so $a = 2^{\frac{2}{3}}$.

Question 6

By selling some crates of soft drinks for ₦600.00, a dealer makes a profit of 50%. How much did the dealer pay for the drinks?

Answer: A

Let the cost price be C . A 50% profit means $600 = 1.5C$. Therefore $C = \frac{600}{1.5} = 400$, so the dealer paid ₦400.00.

Question 7

Find the n th term U_n of the arithmetic progression $11, 4, -3, \dots$.

Answer: C

The common difference is $d = 4 - 11 = -7$. Thus

$$U_n = a + (n - 1)d = 11 + (n - 1)(-7) = 18 - 7n.$$

Question 8

If $R = \{2, 4, 6, 7\}$ and $S = \{1, 2, 4, 8\}$, find $R \cup S$.

Answer: A

The union contains every element in either set, without repetition: $R \cup S = \{1, 2, 4, 6, 7, 8\}$.

Question 9

If $\frac{16}{9}, x, 1, y$ are in geometric progression, find xy .

Answer: C

The common ratio satisfies $x = \frac{16}{9}r$ and $1 = xr$. This gives $r = \frac{3}{4}$, so $x = \frac{4}{3}$ and $y = \frac{3}{4}$. Hence $xy = 1$.

Question 10

[Diagram required] In the Venn diagram, the shaded portion is represented by which set expression?

Answer: C

The shaded part is the part of R outside P , so it is $P^1 \cap R$. The original diagram is required for independent visual verification.

Question 11

Find the values of x for which $\frac{6x-1}{x^2+4x-5}$ is undefined.

Answer: B

The expression is undefined when the denominator is zero. Factor $x^2 + 4x - 5 = (x + 5)(x - 1)$. Therefore $x = -5$ or $x = 1$.

Question 12

Solve the inequality $\frac{1}{3}(2x - 1) < 5$.

Answer: B

Multiply both sides by 3: $2x - 1 < 15$. Then $2x < 16$, so $x < 8$.

Question 13

Find the smaller value of x for which $x^2 - 3x + 2 = 0$.

Answer: D

Factor $x^2 - 3x + 2 = (x - 1)(x - 2)$. The roots are 1 and 2, so the smaller value is 1.

Question 14

Factorize $x(a - c) + y(c - a)$.

Answer: D

Since $c - a = -(a - c)$, $x(a - c) + y(c - a) = x(a - c) - y(a - c) = (a - c)(x - y)$.

Question 15

If $y \propto \frac{1}{x^2}$ and $x = 3$ when $y = 4$, find y when $x = 2$.

Answer: B

Write $y = \frac{k}{x^2}$. When $x = 3$, $4 = \frac{k}{9}$, so $k = 36$. When $x = 2$, $y = \frac{36}{4} = 9$.

Question 16

Solve $3x^2 + 25x - 18 = 0$.

Answer: B

Factor $3x^2 + 25x - 18 = (3x - 2)(x + 9)$. Hence $x = \frac{2}{3}$ or $x = -9$.

Question 17

Solve $(x + 2)(x - 7) = 0$.

Answer: D

Set each factor equal to zero: $x + 2 = 0 \Rightarrow x = -2$, and $x - 7 = 0 \Rightarrow x = 7$.

Question 18

Solve $x + y = \frac{3}{2}$ and $x - y = \frac{5}{2}$. Use your result to find $2y + x$.

Answer: C

Adding the equations gives $2x = 4$, so $x = 2$. Then $2 + y = \frac{3}{2}$, so $y = -\frac{1}{2}$. Therefore $2y + x = 2\left(-\frac{1}{2}\right) + 2 = 1$.

Question 19

[Graph required] Which inequality is illustrated by the sketch graph? The boundary line passes through $(0, 3)$ and $(1, 0)$, and the region below the line is shaded.

Answer: A

The gradient is $\frac{0-3}{1-0} = -3$, so the line is $y = -3x + 3$. The shaded region is below the line, hence $y \leq -3x + 3$.

Question 20

For the graph of $y = 3x^2 + x - 7$, what is the minimum value of y from the given options?

Answer: A

The exact vertex is at $x = -\frac{b}{2a} = -\frac{1}{6}$. Then $y = 3\left(-\frac{1}{6}\right)^2 - \frac{1}{6} - 7 = -\frac{85}{12} \approx -7.08$. From the graph/options, this is read as -7 .

Question 21

Using the graph of $y = 3x^2 + x - 7$, find the roots of the equation.

Answer: A

The roots are where $3x^2 + x - 7 = 0$. By the quadratic formula, $x = \frac{-1 \pm \sqrt{85}}{6}$, giving approximately 1.4 and -1.7 .

Question 22

The diagonals AC and BD of a rhombus $ABCD$ are 16 cm and 12 cm. Calculate the area.

Answer: A

Area of a rhombus $= \frac{1}{2}d_1d_2 = \frac{1}{2}(16)(12) = 96 \text{ cm}^2$.

Question 23

A cylindrical container, closed at both ends, has radius 7 cm and height 5 cm. Find its total surface area.

Answer: B

Total surface area of a closed cylinder is $2\pi r(r + h)$. Using $\pi = \frac{22}{7}$, $r = 7$, $h = 5$:
 $2 \times \frac{22}{7} \times 7 \times 12 = 528 \text{ cm}^2$.

Question 24

Find the total surface area of a solid circular cone with base radius 3 cm and slant height 4 cm. Take $\pi = \frac{22}{7}$.

Answer: A

Total surface area $= \pi r(r + l) = \frac{22}{7} \times 3 \times (3 + 4) = 66 \text{ cm}^2$.

Question 25

A water tank of height $\frac{1}{2} \text{ m}$ has a square base of side $1\frac{1}{2} \text{ m}$. It is filled from a tanker holding 1500 litres. How many litres are left?

Answer: A

Volume $= 1.5 \times 1.5 \times 0.5 = 1.125 \text{ m}^3$. Since $1 \text{ m}^3 = 1000$ litres, the tank holds 1125 litres. Water left $= 1500 - 1125 = 375$ litres.

Question 26

A sphere has volume $k \text{ cm}^3$ and surface area $k \text{ cm}^2$. Calculate the diameter.

Answer: D

Set volume equal to surface area: $\frac{4}{3}\pi r^3 = 4\pi r^2$. Dividing by $4\pi r^2$ gives $\frac{r}{3} = 1$, so $r = 3 \text{ cm}$.
Diameter $= 2r = 6 \text{ cm}$.

Question 27

[Diagram required] A cone has a frustum with radii **12 cm** and **6 cm**, and frustum slant height **12 cm**. Calculate the full cone slant height.

Answer: A

The similar-cone radii are in the ratio $12 : 6 = 2 : 1$, so their slant heights are also in the ratio $2 : 1$. If the small slant height is s , the full slant height is $2s$, and $2s - s = 12$. Thus $s = 12$ and the full slant height is **24 cm**.

Question 28

The positions of two countries P and Q are $(15^\circ N, 12^\circ E)$ and $(65^\circ N, 12^\circ E)$. What is the difference in latitude?

Answer: A

Both latitudes are north, so subtract: $65^\circ - 15^\circ = 50^\circ$.

Question 29

A 120° sector of a circle of radius **21 cm** is bent to form a cone. What is the base radius of the cone?

Answer: D

The sector arc becomes the cone base circumference. Arc length $= \frac{120}{360} \times 2\pi \times 21 = 14\pi$. Since $2\pi r = 14\pi$, $r = 7$ cm.

Question 30

A cylindrical container, closed at both ends, has radius **7 cm** and height **5 cm**. What is the volume?

Answer: A

Volume $V = \pi r^2 h = \frac{22}{7} \times 7^2 \times 5 = 770 \text{ cm}^3$.

Question 31

If $\log_{10} x = \bar{2}.3675$ and $\log_{10} y = \bar{2}.9738$, find $x + y$, correct to three significant figures.

Answer: A

$\bar{2}.3675 = -2 + 0.3675$, so $x = 10^{-2+0.3675} \approx 0.02331$. Also $y = 10^{-2+0.9738} \approx 0.09414$. Therefore $x + y \approx 0.11745$, which is **0.118** to three significant figures.

Question 32

A sector has angle 108° and radius $3\frac{1}{2}$ cm. Find its perimeter.

Answer: D

Perimeter $=$ arc length $+ 2r$. Arc length $= \frac{108}{360} \times 2 \times \frac{22}{7} \times 3.5 = 6.6$ cm. Thus perimeter $= 6.6 + 7 = 13.6 = 13\frac{3}{5}$ cm.

Question 33

[Diagram required] In the diagram, $AO \perp OB$. Find x .

Answer: D

The right angle contributes 90° . The labelled angles around the point give the equation $9x + 90^\circ = 360^\circ$, so $x = 30^\circ$.

Question 34

[Diagram required] In the diagram, O is the centre of the circle and $|BD| = |DC|$. If $\angle DCB = 35^\circ$, find $\angle BAO$.

Answer: C

Using $|BD| = |DC|$, triangle BDC is isosceles. Combining the equal angles with the circle-centre angle facts in the diagram gives $\angle BAO = 20^\circ$.

Question 35

Which of the following angles is an exterior angle of a regular polygon?

Answer: C

An exterior angle of a regular polygon must divide 360° exactly. Only 72° works because $360 \div 72 = 5$.

Question 36

The locus of a point equidistant from two fixed points is the:

Answer: D

All points equidistant from two fixed points lie on the perpendicular bisector of the segment joining those points.

Question 37

[Diagram required] In the diagram, $|AB| = 12 \text{ cm}$, $|AE| = 8 \text{ cm}$, $|DC| = 9 \text{ cm}$, and the indicated lines are parallel. Find $|EC|$.

Answer: A

By similarity from the parallel lines, $\frac{EC}{AE} = \frac{DC}{AB} = \frac{9}{12}$. Hence $EC = 8 \times \frac{9}{12} = 6 \text{ cm}$.

Question 38

[Diagram required] In the diagram, $PQ \parallel TU$, $\angle PQR = 50^\circ$, $\angle QRS = 86^\circ$, and $\angle STU = 64^\circ$. Find x .

Answer: A

Using corresponding and co-interior angles on the parallel lines, the angle chase gives $x = 100^\circ$.

Question 39

[Diagram required] In the diagram, $AB \parallel DC$, and the bisectors of $\angle BAC$ and $\angle ACD$ meet at E . Find $\angle AEC$.

Answer: D

The two angles being bisected are supplementary because of the parallel lines. The bisectors of supplementary angles meet at a right angle, so $\angle AEC = 90^\circ$.

Question 40

[Diagram required] In the diagram, $AB \parallel CD$. What is the size of x ?

Answer: A

Using corresponding angles and the triangle angle sum in the diagram gives $x = 93^\circ$.

Question 41

[Diagram required] In the diagram, find $b + c$.

Answer: C

The diagram forms a cyclic quadrilateral, so the opposite angles b and c are supplementary. Therefore $b + c = 180^\circ$.

Question 42

In $\triangle ABC$, side BC is produced to D . If $|AB| = |AC|$ and $\angle BAC = 50^\circ$, find $\angle ACD$.

Answer: A

Since $|AB| = |AC|$, the base angles are equal: $\frac{180^\circ - 50^\circ}{2} = 65^\circ$. The exterior angle $\angle ACD = 180^\circ - 65^\circ = 115^\circ$.

Question 43

In right triangle PQR , $\angle PRQ = 90^\circ$, $\angle QPR = 30^\circ$, and $|PQ| = 10$ cm. Find $|QR|$.

Answer: C

PQ is the hypotenuse. The side opposite 30° in a right triangle is half the hypotenuse, so $QR = \frac{10}{2} = 5$ cm.

Question 44

Find the mode of 8, 10, 9, 9, 10, 8, 11, 8, 10, 9, 8, 14.

Answer: A

The mode is the value occurring most often. The number 8 appears four times, more than any other value.

Question 45

The bearings of Q and R from P are 030° and 120° . If $|PQ| = 12$ m and $|PR| = 5$ m, find $|QR|$.

Answer: B

The included angle is $120^\circ - 30^\circ = 90^\circ$. Thus $QR = \sqrt{12^2 + 5^2} = 13$ m.

Question 46

Without using tables, evaluate $\frac{\sin 20^\circ}{\cos 70^\circ} + \frac{\cos 25^\circ}{\sin 65^\circ}$.

Answer: A

Since $\sin 20^\circ = \cos 70^\circ$, the first fraction is 1. Since $\cos 25^\circ = \sin 65^\circ$, the second fraction is also 1. Sum = 2.

Question 47

The mean of 20 observations is 4. If the largest observation is 23, find the mean of the remaining observations.

Answer: D

Total = $20 \times 4 = 80$. Removing 23 leaves 57 for 19 observations. New mean = $\frac{57}{19} = 3$.

Question 48

Two fair dice are tossed once. Find the probability that the sum is at least 10.

Answer: A

Favourable ordered pairs are (4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6), so there are 6 outcomes out of 36. Probability = $\frac{6}{36} = \frac{1}{6}$.

Question 49

Find the median of 2.64, 2.50, 2.72, 2.91, 2.35.

Answer: B

In ascending order: 2.35, 2.50, 2.64, 2.72, 2.91. The middle value is 2.64.

Question 50

A box contains 2 red, 6 white, and 5 black balls. A ball is selected at random. What is the probability that it is black?

Answer: A

There are $2 + 6 + 5 = 13$ balls, and 5 are black. Therefore the probability is $\frac{5}{13}$.

Question 51

PART ONE. (a) Factorize $px - 2qx - 4qy + 2py$. (b) Given $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $P = \{1, 2, 4, 6, 10\}$, and $Q = \{2, 3, 6, 9\}$, show that $(P \cup Q)^1 = P^1 \cap Q^1$.

Answer: (a) $(p - 2q)(x + 2y)$. (b) Both sides equal $\{5, 7, 8\}$.

(a) Group terms: $px - 2qx - 4qy + 2py = x(p - 2q) + 2y(p - 2q) = (p - 2q)(x + 2y)$. (b) $P \cup Q = \{1, 2, 3, 4, 6, 9, 10\}$, so $(P \cup Q)^1 = \{5, 7, 8\}$. Also $P^1 = \{3, 5, 7, 8, 9\}$ and $Q^1 = \{1, 4, 5, 7, 8, 10\}$, so $P^1 \cap Q^1 = \{5, 7, 8\}$.

Question 52

(a) y is partly constant and partly varies inversely as the square of x . Write the relationship between x and y . (b) When $x = 1$, $y = 11$, and when $x = 2$, $y = 5$. Find y when $x = 4$.

Answer: (a) $y = a + \frac{b}{x^2}$. (b) $y = \frac{7}{2}$.

(a) $y = a + \frac{b}{x^2}$, where a and b are constants. (b) From $x = 1$, $a + b = 11$. From $x = 2$, $a + \frac{b}{4} = 5$. Subtracting gives $\frac{3b}{4} = 6$, so $b = 8$, and $a = 3$. When $x = 4$, $y = 3 + \frac{8}{16} = \frac{7}{2}$.

Question 53

(a) In the diagram, $PQSR$ and $SRYZ$ are parallelograms, and $PQYZ$ is a straight line. If $|QY| = 2 \text{ cm}$ and $|RS| = 3 \text{ cm}$, find $|PZ|$. (b) Towns P and Q lie on latitude $56^\circ N$, with longitudes $25^\circ E$ and $95^\circ E$. Find PQ along the parallel of latitude, correct to the nearest km. Take $R = 6400 \text{ km}$ and $\pi = \frac{22}{7}$.

Answer: (a) 8 cm. (b) 4374 km.

(a) In parallelogram $PQSR$, $PQ = RS = 3 \text{ cm}$. In parallelogram $SRYZ$, $YZ = RS = 3 \text{ cm}$. Hence $PZ = PQ + QY + YZ = 3 + 2 + 3 = 8 \text{ cm}$. (b) Difference in longitude $= 95^\circ - 25^\circ = 70^\circ$. Distance $= \frac{70}{360} \times 2\pi R \cos 56^\circ \approx 4374 \text{ km}$.

Question 54

(a) A pack of 52 cards is shuffled and one card is drawn. Find the probability that it is either a five or a red nine. (b) The bearing of Q from P is 150° , and the bearing of R from Q is 060° . If $|PQ| = 5 \text{ m}$ and $|QR| = 3 \text{ m}$, find the bearing of R from P , correct to the nearest degree.

Answer: (a) $\frac{3}{26}$. (b) 119° .

(a) There are 4 fives and 2 red nines, so favourable cards = 6. Probability $= \frac{6}{52} = \frac{3}{26}$. (b) Components from P to Q : east $= 5 \sin 150^\circ = 2.5$, north $= 5 \cos 150^\circ = -4.330$. Components from Q to R : east $= 3 \sin 60^\circ = 2.598$, north $= 3 \cos 60^\circ = 1.5$. Thus from P to R : east $= 5.098$, north $= -2.830$. Bearing $\approx 119^\circ$.

Question 55

The table shows marks scored by 32 students. Marks: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Frequencies: 2, 3, 4, 4, 4, 4, 5, 3, 2, 1. (a)(i) Find the mean. (ii) Find the median. (iii) Find the mode. (b) Find the percentage who scored at least 8 marks.

Answer: Mean = 5.22, median = 5, mode = 7, percentage = 18.75%.

Total frequency = 32. $\sum fx = 167$, so mean = $\frac{167}{32} = 5.21875 \approx 5.22$. The 16th and 17th observations are both 5, so median = 5. The highest frequency is 5, at mark 7, so mode = 7. At least 8 marks: $3 + 2 + 1 = 6$. Percentage = $\frac{6}{32} \times 100 = 18.75\%$.

Question 56

PART TWO. (a)(i) Using mathematical tables, find $2 \sin 63.35^\circ$. (ii) Find $\log(\cos 44.74^\circ)$. (iii) Find k given that $\log k - \log(k - 2) = \log 5$. (b) Use logarithm tables to evaluate $\frac{3.68^2 \times 6.705}{\sqrt{0.3581}}$.

Answer: (a)(i) 1.788 approx. (ii) -0.1486 approx. (iii) $k = \frac{5}{2}$. (b) 151.7 approx.

(a)(i) $\sin 63.35^\circ \approx 0.8938$, so $2 \sin 63.35^\circ \approx 1.788$. (ii) $\cos 44.74^\circ \approx 0.7103$, so $\log(\cos 44.74^\circ) \approx -0.1486$. (iii) $\log \frac{k}{k-2} = \log 5$, so $\frac{k}{k-2} = 5$. Hence $k = 5k - 10$, $4k = 10$, and $k = \frac{5}{2}$. (b) $\frac{3.68^2 \times 6.705}{\sqrt{0.3581}} \approx 151.7$.

Question 57

(a) Given $p = x + ym^3$, find m in terms of p, x, y . (b) By completing the square, find the roots of $x^2 - 6x + 7 = 0$ to 1 decimal place. (c) The product of two consecutive positive odd numbers is 195. Form and solve a quadratic equation to find the numbers.

Answer: (a) $m = \sqrt[3]{\frac{p-x}{y}}$. (b) 1.6 and 4.4. (c) 13 and 15.

(a) $p = x + ym^3 \Rightarrow ym^3 = p - x$, so $m = \sqrt[3]{\frac{p-x}{y}}$. (b) $x^2 - 6x + 7 = 0 \Rightarrow (x - 3)^2 - 2 = 0$, so $x = 3 \pm \sqrt{2} \approx 1.6, 4.4$. (c) Let the smaller odd number be n . Then $n(n + 2) = 195$, so $n^2 + 2n - 195 = 0 = (n - 13)(n + 15)$. Hence the numbers are 13 and 15.

Question 58

(a) Complete the table for $y = 2 \cos 2x - 1$ for $x = 0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ, 180^\circ$. (b) Draw the graph for $0^\circ \leq x \leq 180^\circ$. (c) On the same axes, draw $y = \frac{1}{180}(x - 360)$. (d) Use the graphs to find x for which $2 \cos 2x + \frac{1}{2} = 0$.

Answer: Table: 1, 0, -2, -3, -2, 0, 1. Intersections: $x \approx 52.2^\circ$ and 127.8° .

Substitution gives $y = 1, 0, -2, -3, -2, 0, 1$. For part (d), solve $2 \cos 2x + \frac{1}{2} = 0$, so $\cos 2x = -\frac{1}{4}$. In $0^\circ \leq x \leq 180^\circ$, $2x \approx 104.48^\circ$ or 255.52° . Therefore $x \approx 52.2^\circ$ or 127.8° .

Question 59

(a) Using ruler and compasses only, construct $\triangle ABC$ with $|AB| = 7.5 \text{ cm}$, $|BC| = 8.1 \text{ cm}$, and $\angle ABC = 105^\circ$. (b) Locate D on BC such that $|BD| : |DC| = 3 : 2$. (c) Through D , construct a line perpendicular to BC . (d) If the line meets AC at P , measure $|BP|$.

Answer: $|BP| \approx 5.4 \text{ cm}$.

Draw $BC = 8.1 \text{ cm}$. At B , construct 105° , mark A on the ray with $AB = 7.5 \text{ cm}$, and join A to C . Divide BC in the ratio $3 : 2$: $BD = \frac{3}{5} \times 8.1 = 4.86 \text{ cm}$. Draw the perpendicular to BC through D , meeting AC at P . Measurement gives $|BP| \approx 5.4 \text{ cm}$.

Question 60

(a)(i) A man travels from X on bearing 060° to Y , 20 km away. From Y , he travels to Z on bearing 195° . If Z is directly east of X , find YZ to 3 s.f. (ii) Find ZX to 3 s.f. (b)(i) An aircraft flies due south from latitude $36^\circ N$ to $36^\circ S$ on the same longitude. Find the distance travelled to 3 s.f. (ii) If its speed is 800 km/h , find the time to the nearest hour. Take $\pi = \frac{22}{7}$, $R = 6400 \text{ km}$.

Answer: (a)(i) $YZ = 10.4 \text{ km}$. (ii) $ZX = 14.6 \text{ km}$. (b)(i) $8.05 \times 10^3 \text{ km}$. (ii) 10 hours.

(a) From X to Y : north component $= 20 \cos 60^\circ = 10$, east component $= 20 \sin 60^\circ = 17.32$. Since Z is directly east of X , $10 + YZ \cos 195^\circ = 0$, so $YZ = \frac{-10}{\cos 195^\circ} \approx 10.4 \text{ km}$. Easting of Z is $17.32 + 10.35 \sin 195^\circ \approx 14.6 \text{ km}$, so $ZX \approx 14.6 \text{ km}$. (b) Latitude difference $= 72^\circ$. Distance $= \frac{72}{360} \times 2 \times \frac{22}{7} \times 6400 \approx 8045.7 \text{ km} = 8.05 \times 10^3 \text{ km}$. Time $= 8045.7/800 \approx 10$ hours.

Question 61

The table shows scores of 2000 candidates. Percentage marks:

11 – 20, 21 – 30, 31 – 40, 41 – 50, 51 – 60, 61 – 70, 71 – 80, 81 – 90. Frequencies:

68, 184, 294, 402, 480, 310, 164, 98. Prepare a cumulative frequency table and draw the cumulative frequency curve.

Answer: Cumulative frequencies: 68, 252, 546, 948, 1428, 1738, 1902, 2000.

Add the frequencies successively. The cumulative frequencies are

68, 252, 546, 948, 1428, 1738, 1902, 2000. For the ogive, plot cumulative frequency against the upper class boundaries 20.5, 30.5, 40.5, 50.5, 60.5, 70.5, 80.5, 90.5, starting at (10.5, 0), then join the points smoothly.